



Tutor Jolly School

Name: _____

Grade: _____

Class: _____

Class No: _____

HUMAN NUTRITION AND DIGESTION

Section A: Multiple Choice Questions

Select the most appropriate answer for each question. Each question is worth 1 point.

1. Which class of food molecules is chemically digested by the enzyme amylase?

[1Pts]

A. Proteins

B. Fats

C. Starch

D. Vitamins

2. Refer to the diagram. Which organ produces a juice that neutralizes stomach acid and contains multiple digestive enzymes?

[1Pts]

A. Organ A (Liver)

B. Organ B (Stomach)

C. Organ C (Pancreas)

D. Organ D (Large Intestine)

3. What is the primary function of peristalsis in the alimentary canal?

[1Pts]

A. To break down food chemically

B. To absorb nutrients into the blood

C. To push food along the gut

D. To emulsify fats

4. A student tests a food sample with Benedict's solution and heats it. The solution turns brick-red. What is present?

[1Pts]

A. Starch

B. Reducing sugar

C. Protein

D. Lipid

5. The diagram shows a cross-section of a villus. What is the function of the structure labeled X (the central vessel)?

[1Pts]

A. Transport of glucose

B. Transport of amino acids

C. Transport of fatty acids and glycerol

D. Secretion of mucus

6. Which of the following best describes the process of emulsification?

[1Pts]

A. Breaking large fat droplets into smaller droplets to increase surface area

B. Breaking down fat molecules into fatty acids and glycerol chemically

C. Absorption of fats into the lacteal

D. Storage of fats in adipose tissue

7. A deficiency in Vitamin D and Calcium in a growing child is most likely to cause which condition?

[1Pts]

A. Scurvy

B. Anemia

C. Rickets

D. Kwashiorkor

8. In which part of the alimentary canal is water primarily absorbed from undigested food?

[1Pts]

A. Stomach

B. Small intestine

C. Large intestine

D. Rectum

9. Which graph correctly represents the effect of pH on the activity of the enzyme pepsin found in the stomach?

[1Pts]

A. Graph A (Peak at pH 2)

B. Graph B (Peak at pH 7)

C. Graph C (Peak at pH 9)

D. Graph D (Linear increase)

10. What is the correct sequence of processes in human nutrition?

[1Pts]

A. Digestion -> Ingestion -> Absorption -> Assimilation -> Egestion

B. Ingestion -> Digestion -> Absorption -> Assimilation -> Egestion

C. Ingestion -> Absorption -> Digestion -> Egestion -> Assimilation

D. Assimilation -> Digestion -> Ingestion -> Absorption -> Egestion

Section B: Structured Questions

Answer all questions in the spaces provided. Show your working for calculations.

11. The diagram below shows the human teeth arrangement in the lower jaw.

[12 Pts]

- (a) Identify the types of teeth labeled P and Q and state the function of each. (4 marks)
- (b) Describe the cause of dental decay (caries) with reference to bacteria and sugar. (4 marks)
- (c) Explain why mechanical digestion by teeth is important for the efficiency of chemical digestion. (4 marks)

12. The small intestine is the main site of absorption.

[14 Pts]

- (a) Describe three structural adaptations of the small intestine that increase its surface area for absorption. (6 marks)
- (b) Explain the mechanism by which glucose is absorbed into the epithelial cells of the villi against a concentration gradient. (4 marks)
- (c) After absorption, glucose is transported to the liver via a specific blood vessel. Name this blood vessel and explain the role of the liver in regulating blood glucose levels immediately after a meal. (4 marks)

13. A student performed an experiment to determine the energy content of a peanut using the apparatus shown below.

- (a) Using the formula: $\text{Energy (J)} = \text{Mass of water (g)} \times 4.2 \text{ (J/g}^\circ\text{C)} \times \text{Temperature rise (}^\circ\text{C)}$, calculate the energy released if 20g of water was heated from 20°C to 60°C. Show your working. (4 marks)
- (b) The experimental value is often lower than the theoretical energy value of the peanut. Suggest two reasons for this discrepancy based on the setup shown. (4 marks)
- (c) A diet high in fats and sugars can lead to obesity. Discuss the long-term health risks associated with obesity and suggest two dietary changes to manage body mass. (6 marks)
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